EXPERIMENT 1

Design and Prototype of a Cellular System Using Figma

AIM

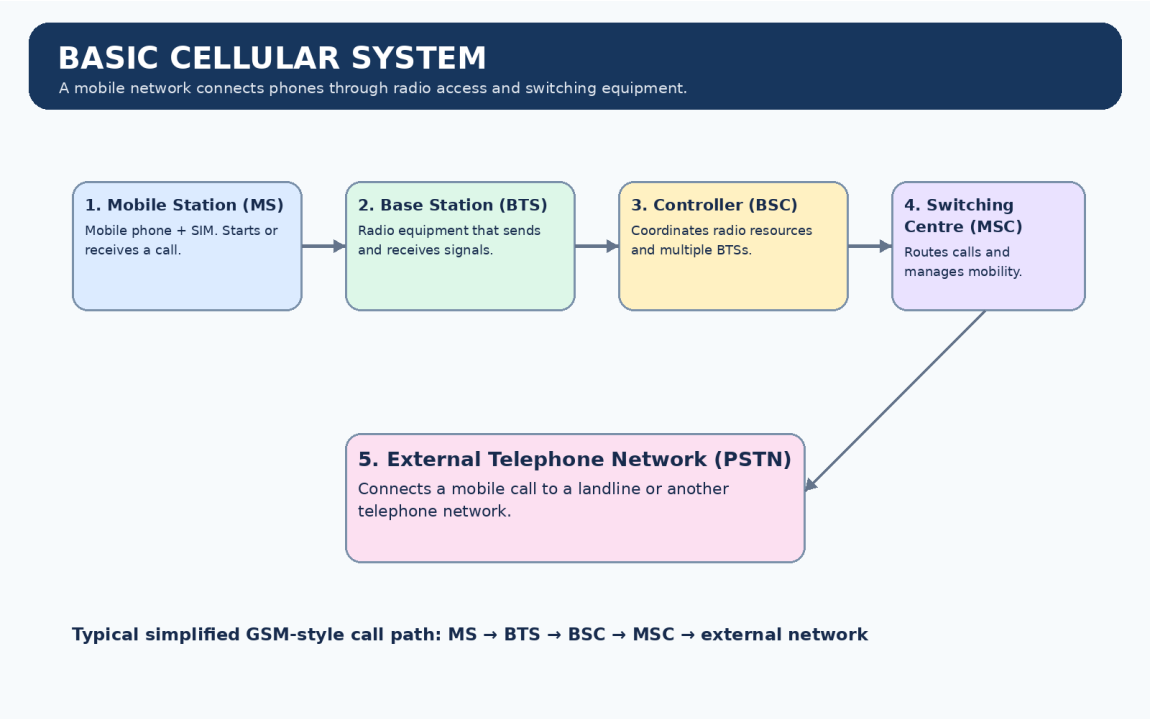
To design the five screens explaining the cellular system (basic cellular system, analog circuit-switched system, components of a cellular system, cellular packet-switched system and digital circuit-switched system) and to connect them in Figma to create a clickable prototype with navigation and return-to-menu links.

PROCEDURE

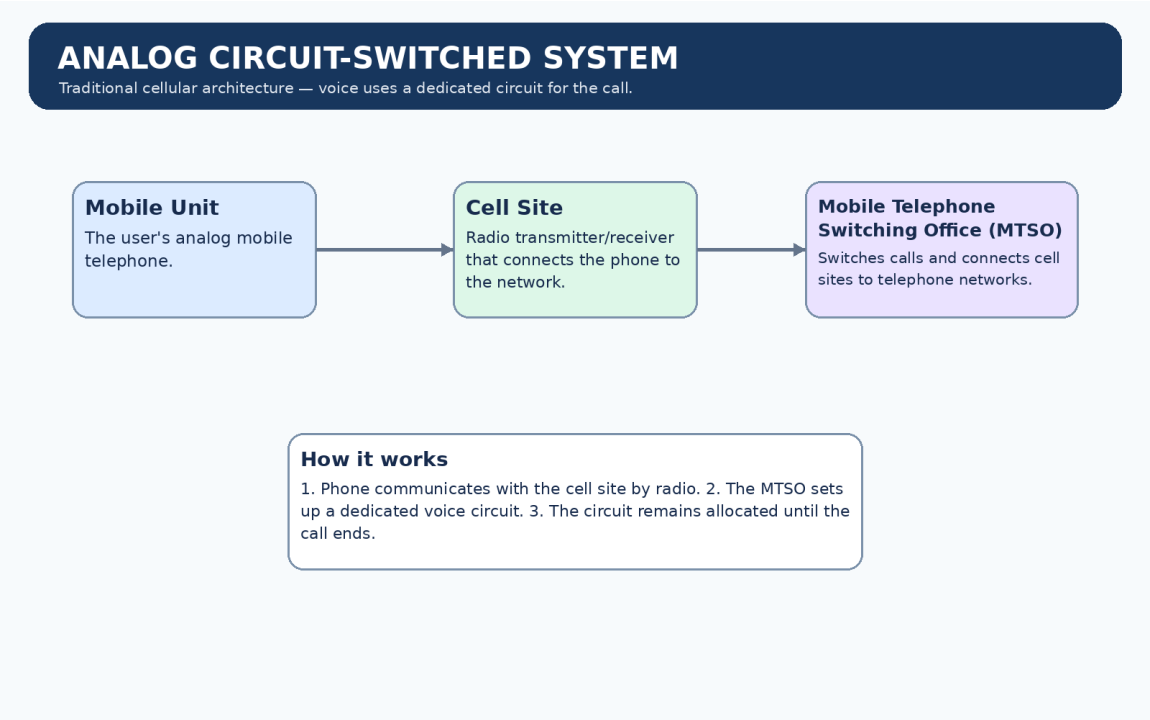
1. Open Figma and create a new Design file.
2. Create five frames (screens) of the same size, one for each topic.
3. Screen 1 - Basic Cellular System: add boxes for MS, BTS, BSC, MSC and the external telephone network (PSTN), joined by arrows.
4. Screen 2 - Analog Circuit-Switched System: add Mobile Unit, Cell Site and MTSO, and a 'How it works' box.
5. Screen 3 - Components of a Cellular System: add MS, BTS, BSC, MSC and External Network with their roles.
6. Screen 4 - Cellular Packet-Switched System: add MS/UE, BTS/Node B, BSC/RNC, SGSN, GGSN and CGF.
7. Screen 5 - Digital Circuit-Switched System: add MS, BTS, BSC, Switching Subsystem, HLR/VLR and External Telephone Network.
8. Use consistent colours, fonts and a heading bar on every screen.
9. Switch to the Prototype tab. Treat Screen 1 as the menu and add a hotspot for each topic.
10. Drag a connection from each hotspot to its screen and set the interaction to 'On click' with 'Navigate to'.
11. On Screens 2 to 5, add a 'Back to menu' connection that returns to Screen 1.
12. Press Play (Present) to test every link, then capture the prototype connection map.

DESIGN

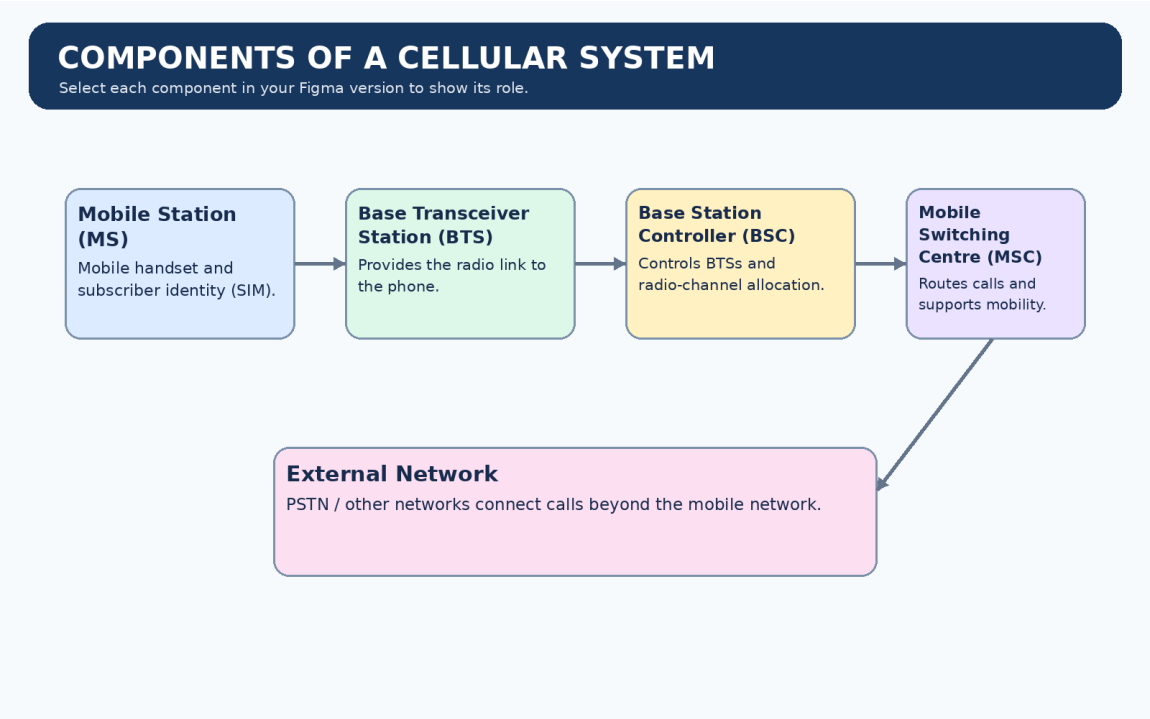
The five designed screens are shown below.



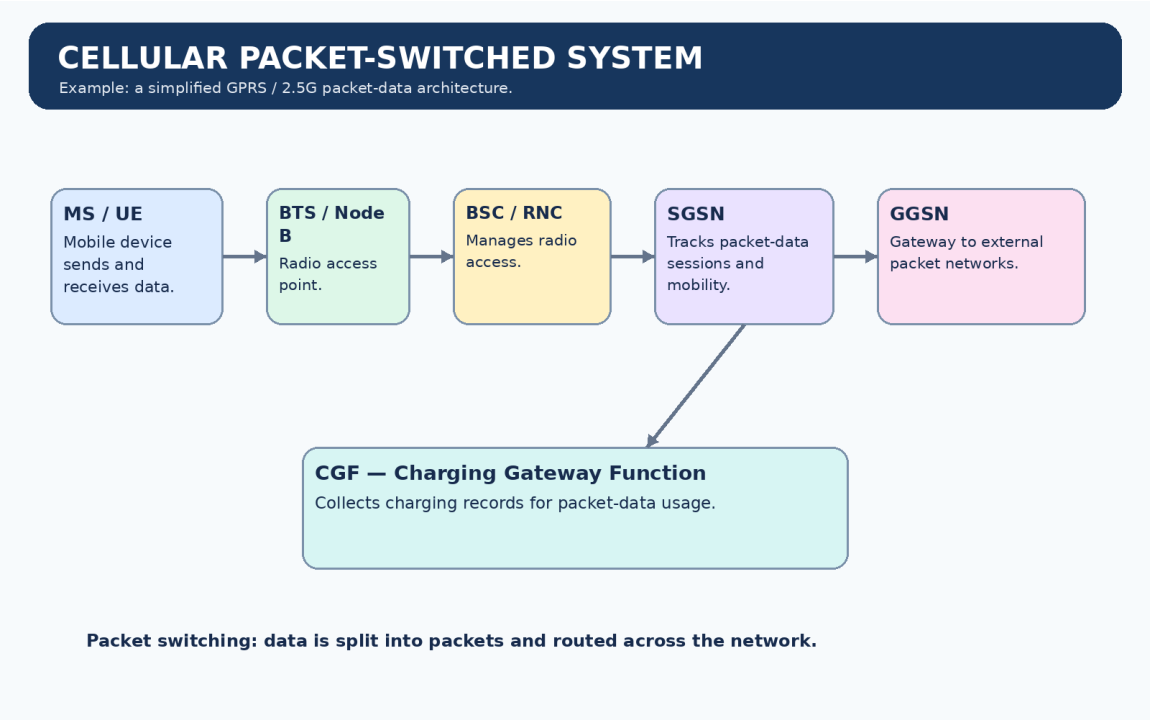
Design 1: Basic Cellular System



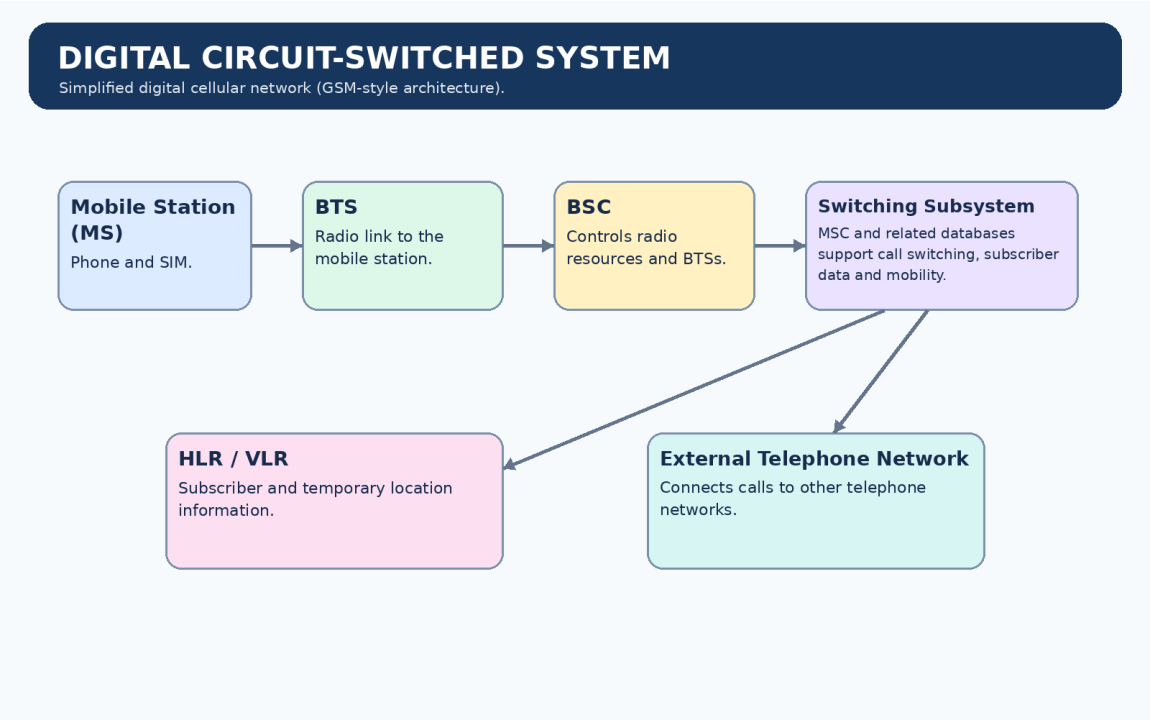
Design 2: Analog Circuit-Switched System



Design 3: Components of a Cellular System



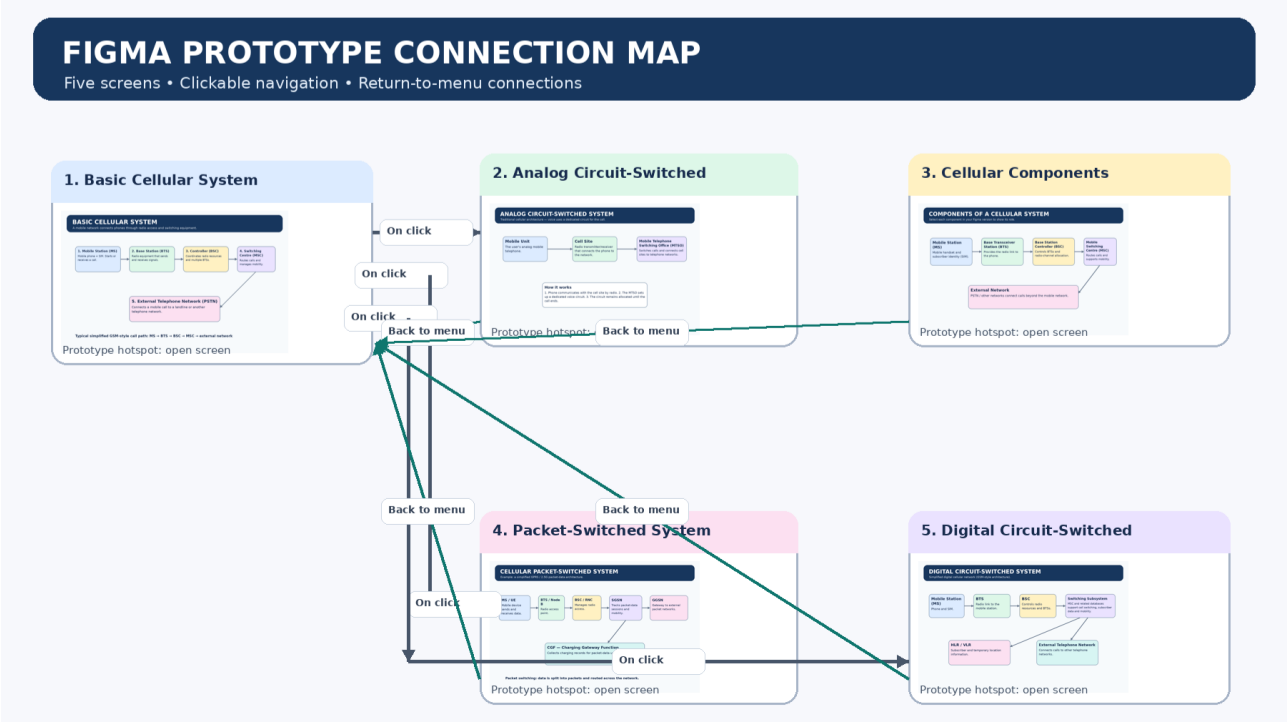
Design 4: Cellular Packet-Switched System



Design 5: Digital Circuit-Switched System

PROTOTYPE

Figma prototype connection map showing the five screens, the 'On click' navigation and the 'Back to menu' connections.



Design 6: Figma Prototype Connection Map

RESULT

The five cellular-system screens were designed in Figma and linked into an interactive prototype. Clicking an item on the menu screen opens the matching screen, and the 'Back to menu' link returns to the menu. The prototype therefore works correctly, and the experiment was completed successfully.